

Companion XXXVII: Disk Dynamics and Rotation Curves in the Relaxation-Driven Cyclic Cosmology

Michael Lehmann

`mi.lehmann@gmx.de`

April 2026

Abstract

The rotation curves of disk galaxies provide one of the most direct observational probes of gravitational dynamics on kiloparsec scales. Their characteristic flatness is traditionally interpreted as evidence for extended dark matter halos. In the Relaxation-Driven Cyclic Cosmology (RDCC), the scale-dependent evolution of the gravitational potential modifies the internal structure of halos and the distribution of angular momentum in a way that reshapes the dynamics of galactic disks. This Companion develops a continuous description of disk formation, derives the RDCC rotation curve, and explains how the relaxation scale influences the transition between rising, flat, and declining regimes. The resulting framework provides a natural interpretation of observed rotation curves without invoking exotic dark matter microphysics.

1 Introduction

Disk galaxies are among the most dynamically ordered structures in the Universe. Their rotation curves, which remain approximately flat far beyond the luminous extent of the disk, have long been regarded as a cornerstone of the dark matter paradigm. In the standard cosmological model, this flatness arises from the presence of an extended halo whose density profile declines more slowly than the baryonic component.

In RDCC, the gravitational potential evolves differently. The relaxation mechanism suppresses the growth of small-scale modes and enhances the coherence of large-scale modes. This modifies the density structure of halos, the distribution of angular momentum, and the gravitational field experienced by the disk. The rotation curve becomes a direct tracer of the relaxation scale k_{rel} and the temporal structure of the potential.

2 The RDCC Gravitational Potential and Disk Formation

The formation of a galactic disk begins with the collapse of gas within a dark matter halo. The gravitational potential governing this collapse is modified in RDCC according to

$$\Phi_{\text{RDCC}}(k, a) = \Phi(k, a) \left[1 - \chi_{\Phi} \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha} \right]. \quad (1)$$

This modification affects the radial distribution of mass within the halo. Small-scale modes, which would normally steepen the inner density profile, are suppressed. As a result, the inner halo becomes less cuspy, while the outer halo retains its coherence. The disk forms within a gravitational field that is smoother and more extended than in the standard cosmological model.

The angular momentum distribution of the collapsing gas is similarly affected. Modes near the relaxation scale dominate the torque field, leading to a more coherent spin axis and a narrower distribution of specific angular momentum.

3 Derivation of the RDCC Rotation Curve

The circular velocity of a test particle in the disk is given by

$$v_c^2(r) = r \frac{\partial \Phi_{\text{RDCC}}}{\partial r}. \quad (2)$$

To evaluate this expression, we consider the RDCC-modified density profile of the halo. For a spherically symmetric halo, the potential satisfies

$$\frac{\partial \Phi_{\text{RDCC}}}{\partial r} = \frac{GM_{\text{RDCC}}(r)}{r^2}, \quad (3)$$

where $M_{\text{RDCC}}(r)$ is the enclosed mass.

The RDCC mass profile can be approximated as

$$M_{\text{RDCC}}(r) = M(r) \left[1 - \chi_M \left(\frac{1}{k_{\text{rel}} r} \right)^\alpha \right], \quad (4)$$

reflecting the reduced contribution of small-scale modes to the inner density.

Substituting this into the circular velocity yields

$$v_{c,\text{RDCC}}^2(r) = \frac{GM(r)}{r} \left[1 - \chi_M \left(\frac{1}{k_{\text{rel}} r} \right)^\alpha \right]. \quad (5)$$

This expression captures the essential RDCC modification: the rotation curve rises more gently in the inner region and transitions smoothly to a flat profile at intermediate radii.

4 The Flatness of Rotation Curves in RDCC

The flatness of rotation curves arises when the enclosed mass grows approximately linearly with radius. In RDCC, this behavior emerges naturally from the interplay between suppressed small-scale modes and coherent large-scale modes. The mass profile becomes

$$M_{\text{RDCC}}(r) \approx \rho_0 r^3 \left[1 - \chi_\rho \left(\frac{1}{k_{\text{rel}} r} \right)^\alpha \right], \quad (6)$$

which yields a circular velocity

$$v_{c,\text{RDCC}}(r) \approx \sqrt{4\pi G \rho_0} r \left[1 - \frac{\chi_\rho}{2} \left(\frac{1}{k_{\text{rel}} r} \right)^\alpha \right]. \quad (7)$$

At radii $r \gg k_{\text{rel}}^{-1}$, the correction term becomes negligible, and the rotation curve approaches a constant value. The flatness is therefore a direct consequence of the relaxation scale, which determines the radius at which the transition occurs.

5 Declining Rotation Curves and the Outer Halo

At sufficiently large radii, the density profile of the halo declines more steeply, and the rotation curve begins to fall. In RDCC, this decline is smoother than in the standard cosmological model because the outer halo retains a higher degree of coherence. The circular velocity in the outer region can be approximated as

$$v_{c,\text{RDCC}}(r) \propto r^{-1/2} \left[1 - \chi_{\text{out}} \left(\frac{1}{k_{\text{rel}} r} \right)^\alpha \right]. \quad (8)$$

This expression predicts a gentle decline in the rotation curve, consistent with observations of extended HI disks.

6 Conclusion

The dynamics of galactic disks in the Relaxation-Driven Cyclic Cosmology reflect the scale-dependent evolution of the gravitational potential. The relaxation mechanism modifies the density structure of halos, the distribution of angular momentum, and the gravitational field experienced by the disk. These effects produce rotation curves that rise gently in the inner region, remain flat over a broad range of radii, and decline smoothly in the outer halo. The present Companion establishes the theoretical foundation for future studies of disk stability, bar formation, and the interplay between baryons and dark matter in RDCC.

References

- [1] Lehmann, M. (2026). Relaxation-Driven Cyclic Cosmology (RDCC): A Minimal CPT-Symmetric Framework with a Single Parameter. Flagship Paper. Zenodo: <https://zenodo.org/records/19744958> (v43.0 or later).
- [2] The complete RDCC companion series (I–LVII) is available at the same Zenodo record above.
- [3] Companion XVI: Boltzmann Code and Modified Hierarchy.
- [4] Companion XXXI: RDCC Halo Model and Non-Linear Structure.
- [5] Companion XXXII–XXXIX: Galaxy–Halo Connection, Cosmic Web, Lensing, RSD, ISW, tSZ, Rotation Curves, CGM.
- [6] Planck Collaboration (2020), *Astron. Astrophys.* 641, A6.
- [7] DESI Collaboration (2024/2025), arXiv: [aktuelle $f\sigma_8$ /BAO-Paper].
- [8] Maiolino et al. (2024), *Nature* (JWST high- z galaxies/quasars).
- [9] Kaiser, N. (1987), *Mon. Not. R. Astron. Soc.* 227, 1 (RSD).
- [10] Bartelmann, M. & Schneider, P. (2001), *Phys. Rep.* 340, 291 (Weak Lensing Review).
- [11] Hu, W. & Okamoto, T. (2002), *Astrophys. J.* 574, 566 (CMB Lensing).
- [12] Lewis, A. & Challinor, A. (2006), *Phys. Rep.* 429, 1 (CMB Lensing Review).
- [13] Navarro, J. F., Frenk, C. S. & White, S. D. M. (1997), *Astrophys. J.* 490, 493 (Halo Profiles).